INFOSYS SPRINGBOARD INTERNSHIP

Data Visualization Domain

Climate Scope: Weather Insights Dashboard

Project Documentation Report

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DOMAIN	:	**Data Visualization**
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Abstract

- This project is about building an interactive weather analytics dashboard using Microsoft Power BI.
- The dataset includes weather records from three countries — Afghanistan, Albania, and Algeria — collected between August 2024 and January 2026.
- The data was first cleaned using Python and then loaded into Power BI for visualization.
- The dashboard has 7 pages. Each page covers a different topic such as temperature, wind, rainfall, regional comparison, extreme weather, and overall weather insights.
- Users can filter the data by country, date, and weather condition using interactive slicers.
- The goal is to make weather data easy to understand and help identify important patterns and trends.

1. Executive Summary

- This project was built as part of the Infosys Springboard Internship under the Data Visualization domain.
- The tool used is Microsoft Power BI.
- The dataset covers weather data from August 2024 to January 2026.
- Three countries are included: Afghanistan, Albania, and Algeria.
- The dashboard has 384 unique location data points.
- It allows users to explore temperature trends, wind patterns, rainfall, regional differences, and extreme weather events.

Key Metrics at a Glance

20.04°C	3.20°C	38.20°C	384
Avg Temperature	Min Temperature	Max Temperature	Location Count

2. Project Overview

2.1 Project Details

Project Title	Climate Scope: Weather Insights Dashboard
Domain	Data Visualization
Tool / Platform	Microsoft Power BI
Data Points	384 unique location-based records
Intern	Gowrisankar V
Mentor	Shanmukha Priya

2.2 Problem Statement

- Weather data is complex and hard to read in raw form.
- There was a need for a simple and visual way to understand weather patterns.

- The goal was to build a dashboard where users can:
- See how temperature changes across time and regions
- Study wind speed and direction patterns
- Compare rainfall levels across different countries
- Find and track extreme weather events
- Get combined insights from temperature, humidity, wind, and rainfall

2.3 Scope

- The project covers the full process — from collecting and cleaning data to building the final dashboard.
- The output is a Power BI report with 7 pages, each focused on a specific weather topic.

3. Data Collection & Preprocessing

3.1 Data Collection

The weather dataset was downloaded from Kaggle.

It contains the following fields:

- Temperature in °C — minimum, maximum, and average
- Humidity (%)
- Wind speed (kph) and wind direction
- Precipitation in mm
- Weather condition (e.g., Sunny, Cloudy, Light Rain)
- Location name and country
- Date of the observation ,etc,…..

3.2 Preprocessing Steps

Python was used with pandas and numpy libraries to clean the data.

The steps followed were:

1. Loaded pandas and numpy libraries.
2. Filled missing number values (temperature, humidity, wind, rainfall) using the median.
3. Filled missing text values (condition, country, location) using the most common value (mode).
4. Saved the cleaned data as a new file for use in Power BI.

3.3 Data Cleaning

- Removed duplicate rows
- Standardized all column names
- Fixed date formats for correct time-series charts
- Verified location names against country data

4. Analysis & Design

4.1 Analysis Methods

- Distribution Analysis — checked how values like temperature and rainfall are spread across countries.

- Correlation Analysis — found relationships between variables like humidity and rainfall.
- Seasonal Trend Analysis — identified how patterns change across months and years.

4.2 Extreme Weather Detection

- Data points above normal ranges were flagged as "highamount".
- All 1,510 records in the dataset were classified as "Normal" during this period.
- This gives a clean baseline for detecting future extreme events.

4.3 Chart Types Used

Chart Type	Used For
Line Chart	Temperature, wind, and rainfall trends over time
Bar Chart	Comparing values across countries
Donut / Pie Chart	Weather condition and wind direction distribution
Scatter Plot	Relationships between temperature, humidity, and wind
Geographic Map	Showing weather patterns on a world map
Heatmap / Matrix	Temperature intensity by country and year
Gauge / KPI Card	Average temperature, rainfall, and wind speed
Waterfall Chart	Rainfall contributions by weather condition

5. Dashboard Development

5.1 Tool Used

- The dashboard was built using Microsoft Power BI Desktop.
- Power BI was chosen because it supports interactive charts, slicers, multi-page navigation, and DAX formulas.

5.2 Dashboard Pages

Page	Page Name	What It Shows
1	**Overview**	KPI cards, map, condition distribution, temperature trend
2	**Temperature Analysis**	Temperature trend, humidity scatter, heatmap, ribbon chart
3	**Wind Analysis**	Wind speed trend, direction chart, speed by condition
4	**Regional Comparison**	Global map, condition distribution, rainfall & temperature by country
5	**Rainfall Analysis**	Rainfall trend, country comparison, waterfall chart
6	**Extreme Weather**	Extreme event count, country-wise extreme weather charts

7	**Weather Insights**	Combined humidity, temperature, wind, and condition trends

5.3 Interactive Features

- Country slicer — filter data by Afghanistan, Albania, or Algeria
- Date range slider — change the time period being viewed
- Weather condition filter — focus on a specific condition like Clear or Heavy Rain
- Temperature range slider — filter by temperature range
- Navigation buttons — go to the next page or back to Overview
- Tooltips — hover over any chart to see more details
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6. Dashboard Pages — Screenshots

6.1 Overview Page

- This is the main page of the dashboard.
- It shows KPI cards for average, minimum, and maximum temperature.
- A map shows temperature by country, and a donut chart shows the weather condition split.
- A line chart shows how temperature changed over time for each country.
- Slicers on the left allow filtering by country, date, and weather condition.

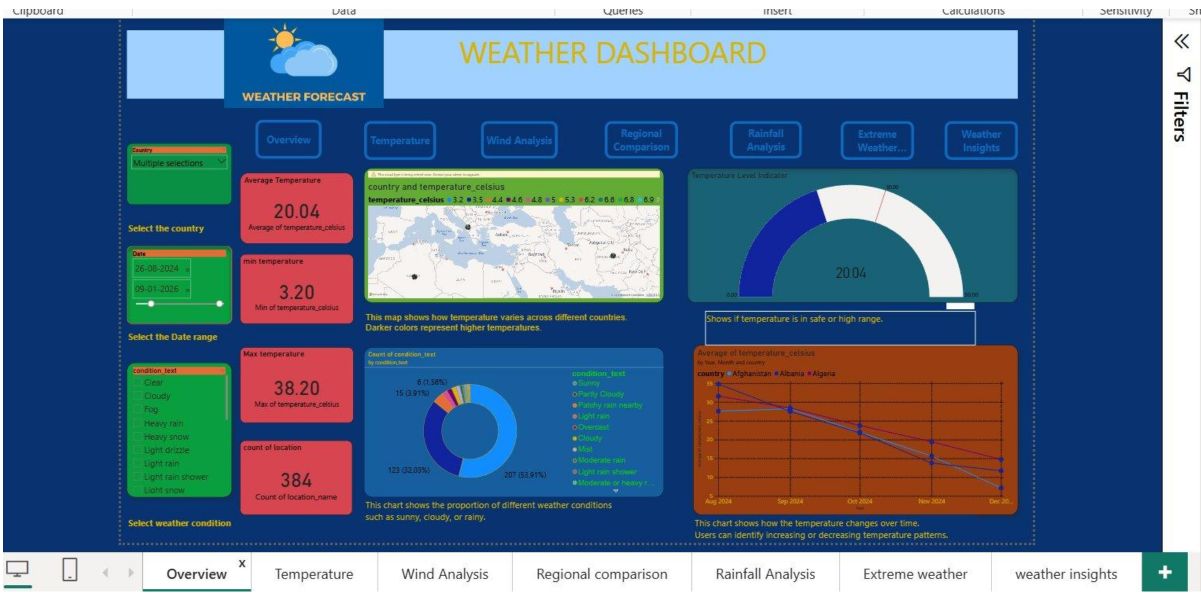

Figure 1: Overview Page

6.2 Temperature Analysis Page

- This page shows temperature data in detail.
- A line chart shows the temperature trend from 2024 to 2026.
- A scatter plot shows the relationship between humidity and temperature.
- A bar chart compares average temperature across countries.
- A heatmap shows temperature totals per country per year.

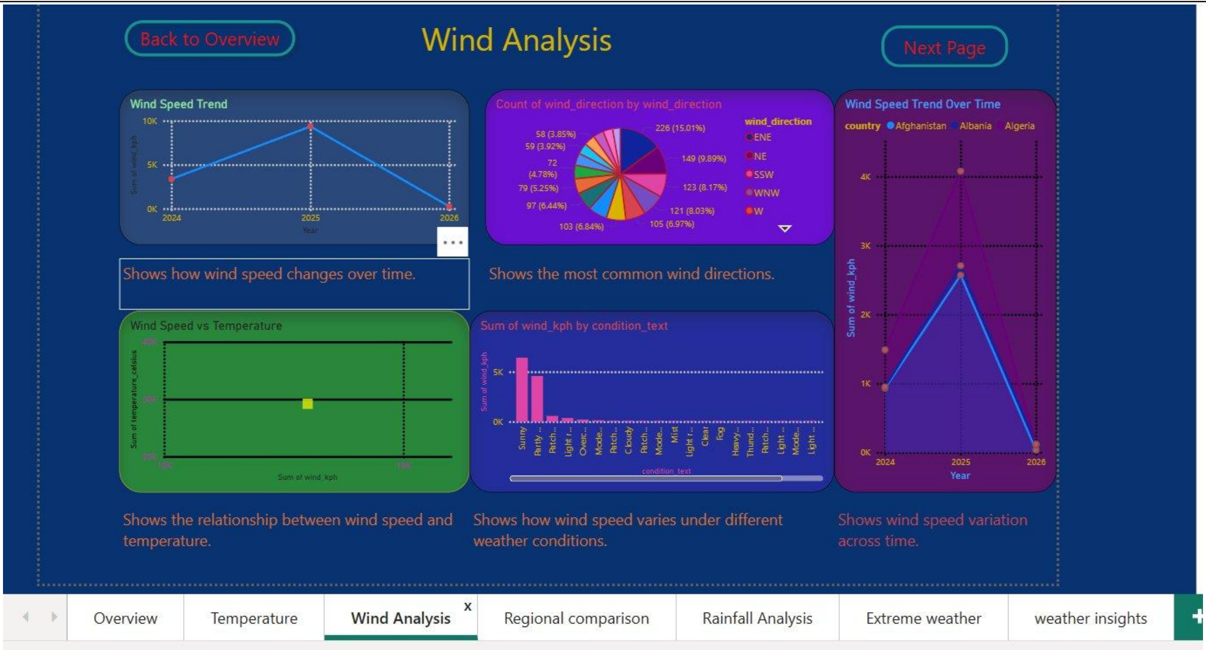

Figure 2: Temperature Analysis Page

Key Finding: Algeria had the highest cumulative temperature (10,244.40). Temperature peaked in 2025 and dropped in 2026.

6.3 Wind Analysis Page

- This page covers wind speed and direction.
- A line chart shows wind speed changed over 2024 to 2026.
- A pie chart shows the most common wind directions — ENE (15.01%) and NE (9.89%).
- A bar chart shows how wind speed varies by weather condition.
- A multi-line chart compares wind speed trends across all three countries.

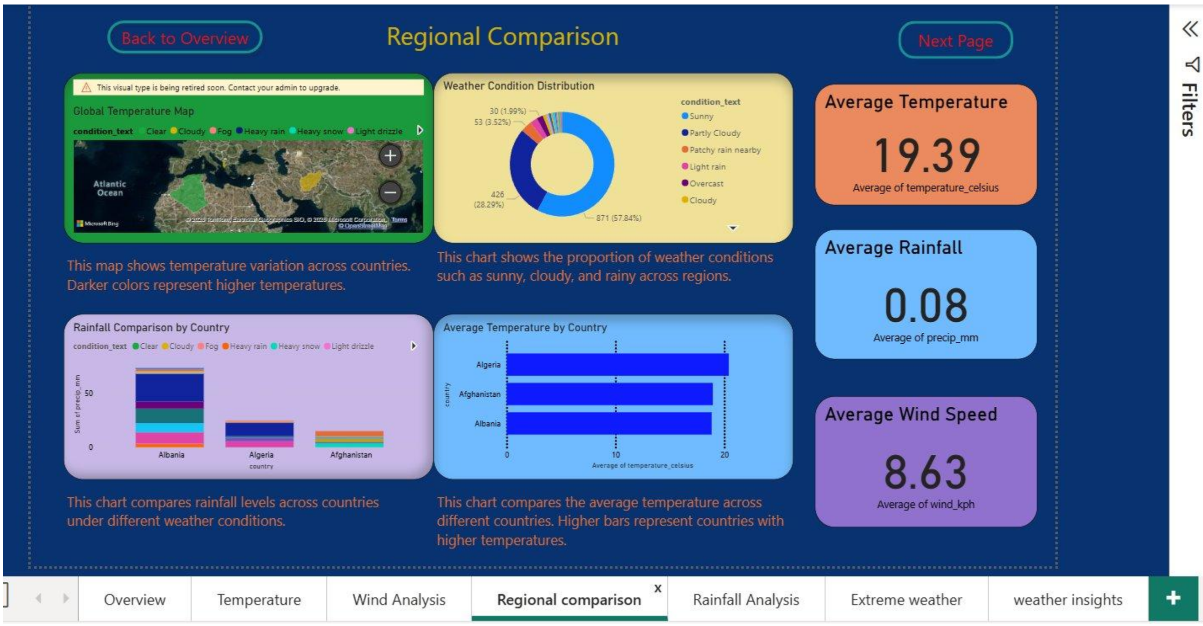

Figure 3: Wind Analysis Page

Key Finding: ENE and NE are the most common wind directions. Wind speed peaked in 2025 and dropped by 2026.

6.4 Regional Comparison Page

- This page compares weather across the three countries.
- A global map shows temperature intensity by location.
- A donut chart shows that 57.84% of records were Sunny and 28.29% were Partly Cloudy.
- Bar charts compare rainfall and average temperature side by side.
- KPI cards show: Average Temperature 19.39°C, Average Rainfall 0.08 mm, Average Wind Speed 8.63 kph.

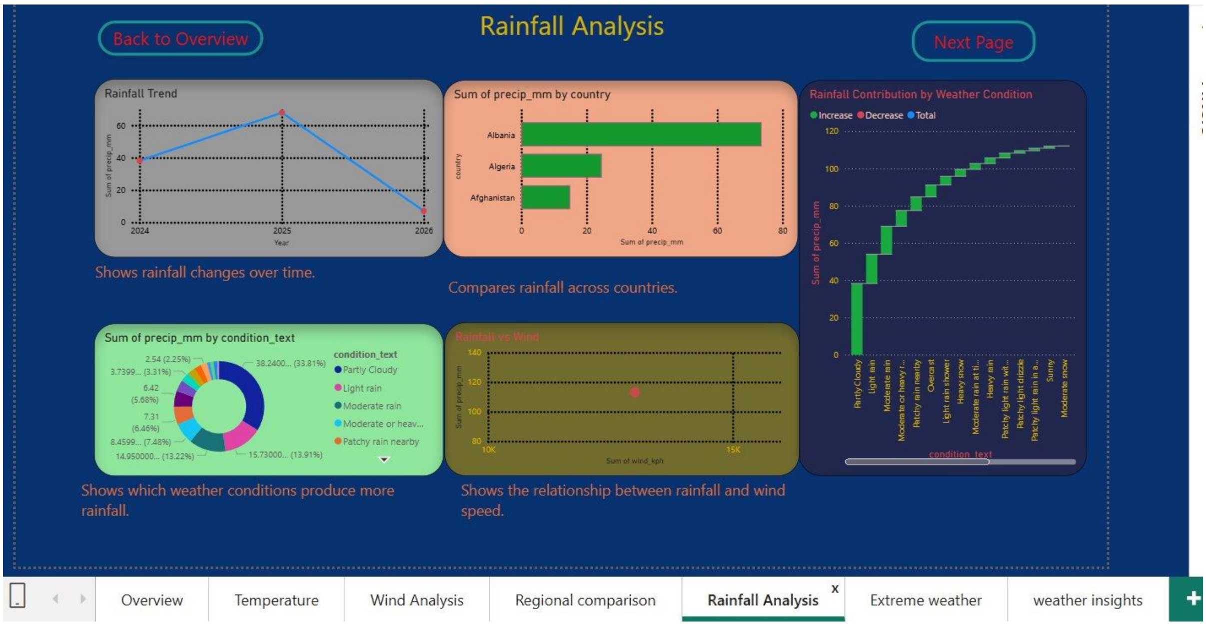

Figure 4: Regional Comparison Page

6.5 Rainfall Analysis Page

- This page shows rainfall patterns across time and countries.
- Rainfall increased from 2024 to mid-2025 and then dropped.
- Albania had the most total rainfall.
- A donut chart shows Partly Cloudy conditions produced the most rainfall (33.81%).
- A waterfall chart shows how each condition contributes to total rainfall.
- A scatter plot shows a weak link between rainfall and wind speed.

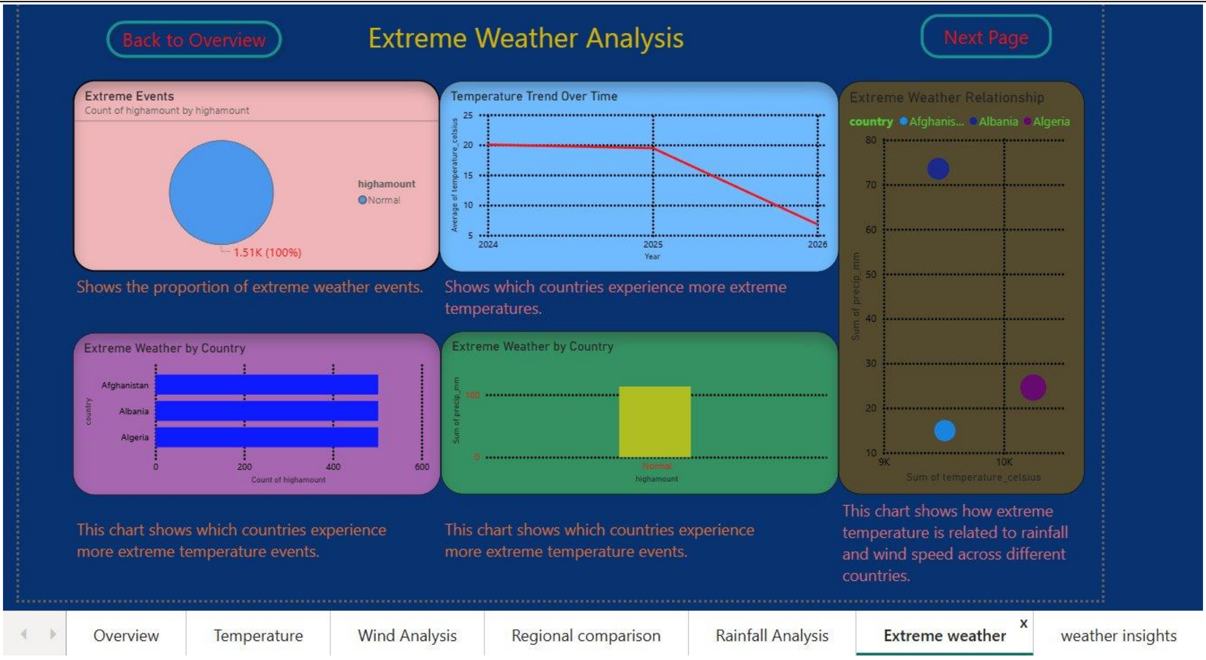

Figure 5: Rainfall Analysis Page

Key Finding: Partly Cloudy conditions contributed the most rainfall. Rainfall and wind speed have a weak positive link.

6.6 Extreme Weather Analysis Page

- This page looks at extreme weather events.
- All 1,510 records were classified as "Normal" — no extreme events were found.
- A temperature trend chart shows average temperature dropped from 20°C to below 10°C by 2026.
- Bar charts compare extreme event counts by country.
- A bubble chart links extreme temperature with rainfall and wind speed.

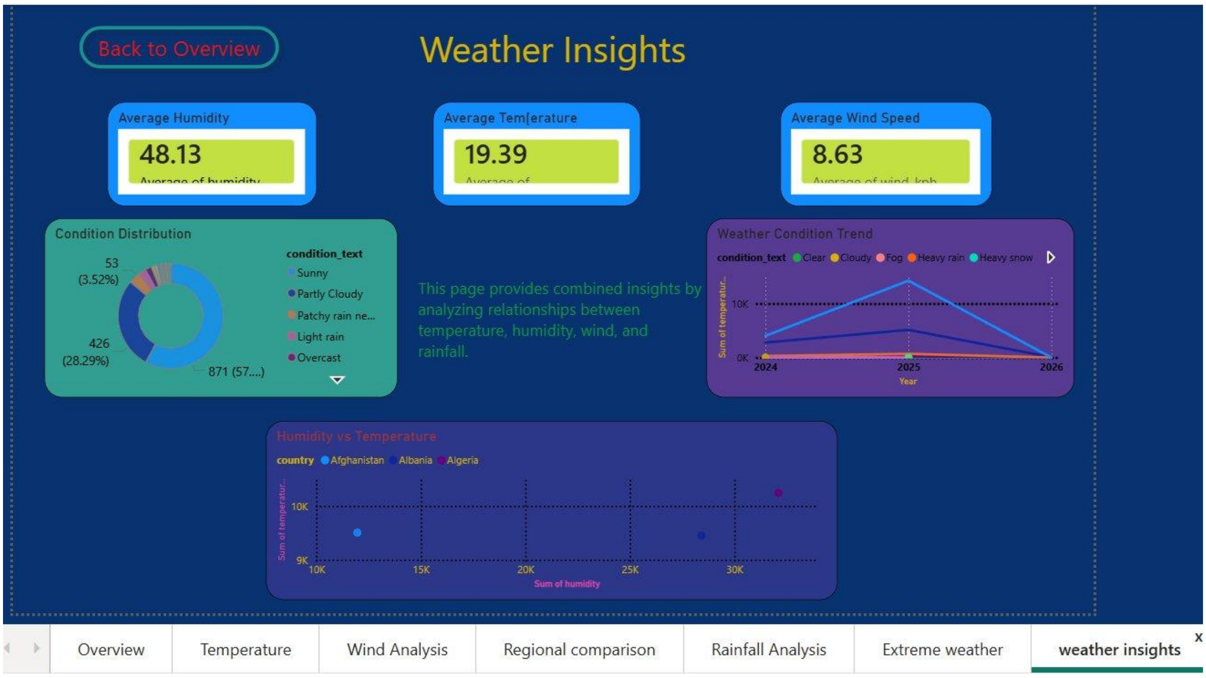

Figure 6: Extreme Weather Analysis Page

6.7 Weather Insights Page

- This page combines all weather data into one view.

- Three KPI cards show: Average Humidity 48.13%, Average Temperature 19.39°C, Average Wind Speed 8.63 kph.
- A donut chart and a trend chart show that Sunny conditions were the most common (57.84%) and peaked in 2025.
- A scatter plot shows the relationship between humidity and temperature for each country.

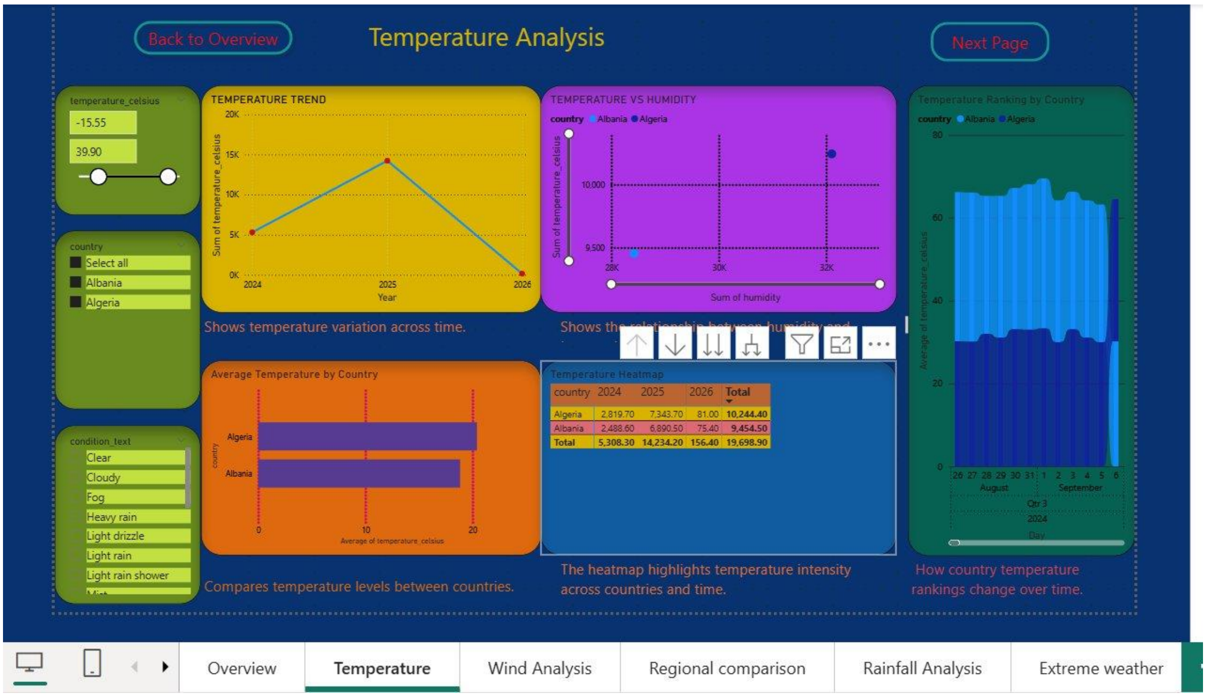

Figure 7: Weather Insights Page

7. Key Insights & Findings

Category	Finding
Temperature	Average temperature is 20.04°C. Algeria is the warmest. Temperature peaked in 2025 and dropped in 2026.
Humidity	Average humidity is 48.13%. Higher humidity in Algeria links to more rainfall.
Wind	Average wind speed is 8.63 kph. ENE and NE are the most common directions. Wind was strongest in 2025.
Rainfall	Average rainfall is 0.08 mm. Albania has the most total rain. Partly Cloudy produced the most rainfall (33.81%).
Conditions	Sunny was the most common condition (57.84%). Light rain and partly cloudy were next.
Extreme Events	No extreme events found. All 1,510 records were Normal.
Regional Gap	Afghanistan is hot but dry. Algeria is hot with more rainfall due to its coastal climate.

8. Project Milestones

Milestone 1 — Data Collection & Preprocessing

- Collected weather data from Kaggle for three countries
- Removed duplicates and fixed column names
- Filled missing numbers with median and missing text with mode
- Saved the cleaned dataset for Power BI

Milestone 2 — Analysis & Design

- Ran statistical analysis on temperature, wind, humidity, and rainfall
- Found seasonal trends across months and years
- Set thresholds to identify extreme weather events
- Planned the dashboard layout and selected chart types for each page

Milestone 3 — Dashboard Development

- Built all 7 pages of the dashboard in Power BI
- Added slicers for country, date, temperature, and weather condition
- Added navigation buttons on every page
- Improved the design with consistent colors, labels, and chart annotations
- Checked all charts to make sure they match the data correctly

9. Challenges & Learnings

9.1 Challenges Faced

- Some columns had missing values and needed to be filled carefully
- Designing 7 pages that are easy to navigate without making the report slow
- Keeping a consistent look across many different chart types
- The map visual in Power BI was flagged as retiring and needed an alternative plan

9.2 Skills Learned

- Full data analytics process — from raw data to dashboard
- Advanced Power BI features — DAX, slicers, bookmarks, navigation, heatmaps
- How to interpret weather data and detect unusual patterns
- Data storytelling — how to guide the viewer through a dashboard
- Python preprocessing with pandas and numpy on real-world data

10. Conclusion & Next Steps

10.1 Conclusion

- The Climate Scope dashboard was successfully completed.
- It covers all major weather topics across 7 pages with interactive filters.
- The project shows the full data visualization process — from data collection and cleaning to analysis and dashboard building.
- It demonstrates skills in Python, Power BI, DAX, and data storytelling.

10.2 Future Enhancements

- Add machine learning models (e.g., ARIMA, LSTM) to predict future weather trends.
- Include more countries to make the analysis more global.
- Connect to a live weather API (e.g., OpenWeatherMap) for real-time data.
- Build an alert system that sends notifications when extreme weather is detected.
- Optimize the dashboard for Power BI mobile view.